PROJECT REPORT

ON

“Solar Dryer”

SUBMITTED BY

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Under Guidance of

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Diploma Course in Information Technology

(As per directives of I Scheme, MSBTE)

Sinhgad Institutes

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SOU. VENUTAI CHAVAN POLYTECHNIC, PUNE - 411041

ACADEMIC YEAR 2023 – 2024

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Date – 02/01/2024

To,

Mr.U.S.Shirshetti
Head of Department
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Sou.Venutai Chavan Polytechnic, Pune

Name of Students-
Ms.Snehal Pawar
Ms.Sakshi Gawali
Ms.Kajal Dhawade
Ms.Prashika Waghmare

To Whom It May Concern

Subject- Sponsorship for Academic Project

Respected Sir,

We, OnBoard Services Pvt Ltd, are pleased to inform you that the Snehal Pawar, Sakshi Gawali, Kajal Dhawade and Prashika Waghmare TY Information Technology Department are approved for the sponsored Project titled **"Solar Dryer"**- in our Company.
We will be sponsoring the students in the form of guidance and finances (If required).

Hope this is in-line of your requirements. You may communicate us on our E-mail ID For any query.
Thanking You,

For OnBoard Services Pvt.Ltd,

ONBOARD SERVICES

Authorized Signature

ABSTRACT

A solar dryer is another technology to harness the solar energy that is used to dry fruits, vegetables, and crops for preservation. Solar dryers are of two types: direct and indirect. In direct solar dryers, the substance that is to be dehydrated is exposed to the sunlight in a vast field. Indirect solar dryers consist of an insulated box coated inside with a black absorption surface, an air inlet and an air outlet, and single- or double-glazed glass. A solar dryer works on the principle of the density differential. The inlet air hole is at the lower side for the entrance of the cold air and the outlet air is at the upper side of the opposite wall. The sunlight coming through the glazing keeps the inner environment warm, which dehydrates the substance. The cold air takes the hot air enriched with moisture content from the box and, because of the difference in the density, the air is ventilated through the hot air outlet.

Lists Of Tables

ACKNOWLEDGEMENTS

“Let Us Make Our Future Now & Let Us Make Our Dreams Tomorrow’s Reality”

I find great pleasure in expressing the gratitude towards the all, which made it possible for me to complete this dissertation successfully.

“**Perseverance, inspiration and motivation plays a key role in the success of any venture very fast”,** I would like to express my deepest and sincerest gratitude to internal project guide **Mr. U. S. Shirshetti** for his dynamic and valuableguidance and keen interest in my dissertation work. I am grateful to him for his constant encouragement in the fulfillment of the dissertation work. He provides me the valuable knowledge by sharing vast experience through plenty of his precious time.

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Snehal Pawar

Sakshi Gawali

Kajal Dhawade

Prashika Waghmare

Lists Of Tables

Sr. No	Title	Page No
1.1	INTRODUCTION & BACKGROUND	2
2.1	PRELIMINARY SURVEY	5
3.1	FEASIBILITY STUDY	9
4.1	PROPOSED METHODOLOGY	17
5.1	ACTION PLAN	19
6.1	CONCLUSION & FUTURE SCOPE	29
7.1	ACTUAL SCREEN LAYOUT	31
8.1	REFERENCE	34

Lists Of Figures

Figure No	Title	Page No
4.1	Solar Panel	11
4.2	Rechargeble Batterry	12
4.3	Fan and Air Pump	13
4.4	Relay Module	14
4.5	Arduino	15

Contents

Chapter - 1 INTRODUCTION & BACKGROUND

1.1 Introduction

1.2 Background

Chapter - 2 PRELIMINARY SURVEY

2.1 Literature Survey

2.2 Problem Definition

2.3 Specification

Chapter - 3 FEASIBILITY STUDY

3.1 Technical Feasibility

3.2 Economical Feasibility

3.3 Operational Feasibility

Chapter - 4 ANALYSIS

4.1 System Analysis

4.2 Data Collection

4.3 Information Gathering tools

4.4 User requirement analysis

Chapter - 5 PROPOSED METHODOLOGY

5.1 Proposed Methodology

5.2 Architecture/Block Diagram

5.3 Action Plan

Chapter - 6 PROGRAMMATIC CODE

Chapter - 7 USER MANUAL

7.1 Installation and Setup

7.2 Input to Application

7.3 Output from Application

Chapter-8 ADVANTAGES AND APPLICATIONS

8.1 Advantages

8.2 Applications

Chapter-9 CONCLUSION AND FUTURE SCOPE

9.1 Conclusion

9.2 Future Scope

Chapter - 10 ACTUAL SCREEN LAYOUT

Chapter - 11 REFERENCES

11.1 Research papers

CHAPTER 1

INTRODUCTION AND BACKGROUND

1.1 INTRODUCTION

Drying of food grains in the fields by exposure to the sun has been very common in India since ancient times. Industrialization in the present century created a demand for controlled drying of many agricultural products such as tobacco, wood, peaches, resins, etc., since such products retain the flavor, quality, and appearance and thus have better sale prospects. The other advantage is that such products could be dried in peak season and made available for consumption throughout the year.Coal, oil, or firewood are usually used in such conventional industrial dryers. The present energy crisis compels one to think of the use of alternate sources of energy and the population free. Abundant and readily available solar energy seems to be the only answer. Solar dryers utilize the sun's heat to remove moisture from fruits, vegetables, herbs, and even meats, extending their shelf life and preventing spoilage. This age-old technique has been practiced for centuries in various forms across different cultures, but with advancements in technology and design, modern solar dryers offer increased efficiency and consistency. The system's goal is to provide high-quality drying while reducing labour costs and increasing productivity. The use of solar power and automation ensures a costeffective and consistent operation, making it an attractive solution for drying various products.

1.2 BACKGROUND

Drying by exposure to the Sun is one of the oldest methods using solar energy, for food preservation, as vegetables, fruits, fish, meat, etc. Already from prehistoric times mankind used solar radiation as the only available thermal energy source to dry and preserve all necessary foodstuffs for winter time, to dry soil bricks for their homes, and animal skins for dressing. Arian (2nd century AD; Translation, 1966) describes that when Nearchus, the admiral of Alexander the Great, was sailing with his fleet across the Macran Coast (today in the border of Iran/Pakistan), they arrived at a place called “Mahi Khoren” which in Persian means fish-eaters (ichthiophagi). They dried large fishes by exposure to the sun and poured the dried fish to flour to produce bread and cakes. Until today, the same drying situation in this region.

The first known drying installation has been found in the South of France and is dated from about 8000 BC. It was a stone-paved surface and was used for drying of crops. Breezes or natural moderate wind velocities were combined with solar radiation to accelerate drying (Kröll and Kast, 1989).

Various other installations around the world have been found dated between the years 7000 and 3000 BC. There were various combined installations, solar radiation natural air circulation, for drying mainly food. In Mesopotamia have been found sites, of solar and air drying colored textile material and for solar air drying of written clay plates. The first, exclusively air-drying installation for crops was found in the Hindu river valley and is dated about 2600 BC (Kröll and Kast, 1989).

CHAPTER 2

LITERATURE SURVEY

2.1 LITERATURE SURVEY

Solar drying is a sustainable and efficient method of drying agricultural products that has been gaining popularity in recent years. Numerous studies have been conducted on solar dryers, and their results have shown that they can provide high-quality drying while reducing labour costs and energy consumption.

One study conducted by Singh and Tiwari (2007) developed a natural convection solar dryer for drying grapes. The system used a solar collector and a drying chamber, and the drying process was controlled using temperature and humidity sensors. The results showed that the solar dryer reduced the drying time and improved the quality of the dried grapes compared to sun drying.

In recent years, the use of microcontrollers such as Arduino has become popular in the development of solar dryers. Arduino microcontrollers provide a cost-effective and easy-to-use platform for controlling and monitoring the drying process.

One study by Srinivasan and Sangeetha (2016) developed a solar dryer using Arduino for drying mushrooms. The system used a temperature and humidity sensor for monitoring the drying process, and a fan for efficient drying. The results showed that the solar dryer reduced the drying time and improved the product quality compared to traditional drying methods.

Overall, the literature review suggests that solar dryers, particularly those incorporating microcontrollers such as Arduino, provide a sustainable and efficient method of drying agricultural products. These systems have the potential to reduce labour costs, improve product quality, and contribute to environmental sustainability.

2.2 Problem Definition

These methods also consume a significant amount of energy, contributing to environmental degradation. As a result, there is a need for modern and efficient drying methods that address these drawbacks. The problem statement for this project is to design and develop a solar dryer using Arduino for the drying of agricultural products. The system should provide high-quality drying while reducing labor costs and increasing productivity.

Objective:

The objectives of the solar dryer using Arduino project are:

1. To design and develop a smart solar dryer using Arduino for the drying of agricultural products.
2. To automate the drying process using an Arduino controller, ensuring consistent and optimal drying conditions.
3. To use solar power as the primary source of energy, making the system cost-effective and environmentally friendly.
4. To incorporate a fan, LED lights, and a relay module for efficient and effective drying.
5. To provide high-quality drying while reducing labour costs and increasing productivity.

2.3 Specification

- **Software Specification**
    1. Operating System: Linux/Windows.
    2. Language: C Programming
    3. Software with version: Android Studio.

- **Hardware Specification**
  1. Arduino Board
  2. Solar Collector
  3. Control System
  4. Solar Panel
  5. Camera

- **Features**
  1. Air Circulation System
  2. Drying Chamber
  3. Ventilation System
  4. Thermal
  5. Control Mechanism
  6. Weather Protection
  7. Modularity and Portability
  8. Energy Efficiency
  9. Low Operating Costs

CHAPTER 3

FEASIBILITY STUDY

Feasibility study

The feasibility study of the "**Solar Dryer**." project covers three major aspects namely Technical, Economical,and Operational.

3.1 Technical Feasibility

A feasibility study is an in-depth examination of the technical feasibility and financial viability of the proposed project. The feasibility study would usually integrate all aspects of the project including computer vision and hand gesture recognition systems, ensuring the seamless translation of real-world hand movements into immersive virtual environments. The incorporation of machine learning models enhances the precision and adaptability of gesture detection, offering an intuitive user experience. The study encompasses all technical aspects, ensuring compatibility with diverse Solar platforms and emphasizing real-time simulation responsiveness with a smooth frame rate.

3.2 Economic Feasibility

The economic feasibility step of business development is that period during which a break- even financial model of the business venture is developed based on all costs associated with taking the product from idea to market and achieving sales sufficient to satisfy debt or investment requirements.

3.3 Operational Feasibility

Operational feasibility is **the measure of how well a proposed system solves the problems.** and takes advantage of the opportunities identified during scope definition and how it satisfies the requirements identified in the requirements analysis phase of system development.

CHAPTER 4

ANALYSIS

4.1 System Analysis

Solar Panel

Solar panels are devices that convert sunlight into electrical energy. They are made up of several solar cells that are connected together and are designed to absorb the photons present in sunlight and convert them into electrical energy through the photovoltaic effect.

Specifications:

1. Efficiency: Efficiency refers to the percentage of sunlight the solar panel can convert into electricity. Higher-efficiency panels produce more power for a given area and are often preferred when space is limited.
2. Dimensions: The physical dimensions of the solar panel, including length, width, and thickness, are important considerations for installation.
3. Weight: The weight of the solar panel is important for structural considerations, especially for rooftop installations or applications where weight is a concern.
4. Temperature Coefficient: The temperature coefficient indicates how the power output of the solar panel decreases with increasing temperature.
5. Maximum Voltage: This is the voltage at which the solar panel operates most efficiently. It's important to match the panel with the inverter and electrical system.

Rechargeable Battery

A battery is an important component of a solar dryer, especially in off-grid or remote areas where access to the power grid is limited or unreliable. The battery serves as a storage device for the energy generated by the solar panels during the day, which can be used to power the dryer during periods of low sunlight or at night.

Specifications:

1. Lens Type: The lens type and focal length influence the camera's focus, zoom capability, and depth of field. Choose a lens with suitable characteristics for capturing clear images of the drying process from the desired distance and angle.

2. Field of View:The field of view determines the area that the camera can capture. Consider the desired coverage area and angle when selecting a camera with an appropriate FOV to ensure comprehensive monitoring of the solar dryer system.

3. Image Sensor Type: Different types of image sensors, such as CMOS (Complementary Metal-Oxide-Semiconductor) or CCD (Charge-Coupled Device), offer varying levels of sensitivity, dynamic range, and noise performance. Choose a camera with an appropriate sensor type based on the specific requirements of your monitoring application.

Fan and Air Pump:

Fans are used to increase the air flow through the dryer tunnel, which can help to speed up the drying process and improve the overall efficiency of the system. Fans can be powered by solar panels or a battery, and may be operated manually or automatically using a thermostat or humidity sensor to control their operation.

Relay Module:

A relay module is an electronic component commonly used in solar dryers to control the flow of electricity to different components. It works as an on/off switch that is controlled by an electrical signal.

Arduino :

Arduino is an open-source electronics platform based on easy-to-use hardware and software. It consists of a microcontroller board and a development environment that can be used to create a wide variety of electronic projects, including solar dryers.

The Arduino board is typically equipped with input/output (I/O) pins that can be used to control and monitor various components of a solar dryer system, such as sensors, relays, and displays. The board can be programmed using the Arduino Integrated Development Environment (IDE), a software platform that provides a simple and user-friendly interface for writing and uploading code to the board. Overall, Arduino is a powerful and flexible tool for designing and implementing solar dryer systems, providing users with a simple and affordable way to control and monitor various components of the system.

4.2 Data Collection

We collected the data from the internet and some reference books. We also got some data professionals who gave us guidance on how to do the project. Mostly weused the knowledge for the project and also collected data from various sources likebooks, Wikipedia, the internet, etc.

4.3 Information Gathering Tools

Internet

The Internet was a good information-gathering tool for us. It contains information about any concept, any person required. That is the reason why information about components as well as implementation of the whole project was easy.

Our own knowledge

We also went through a couple of books and many people, and contributed with our own knowledge to accomplish this particular project.

4.4 User Requirement Analysis

- Arduino Board
- Solar Panel
- Rechargeable Battery
- Temperature Sensor
- Humidity Sensor
- Camera
- Relay Module
- Transistor NPN
- DC Fan
- Resistors
- Zero PCB
- 16x2 LCD

CHAPTER 5

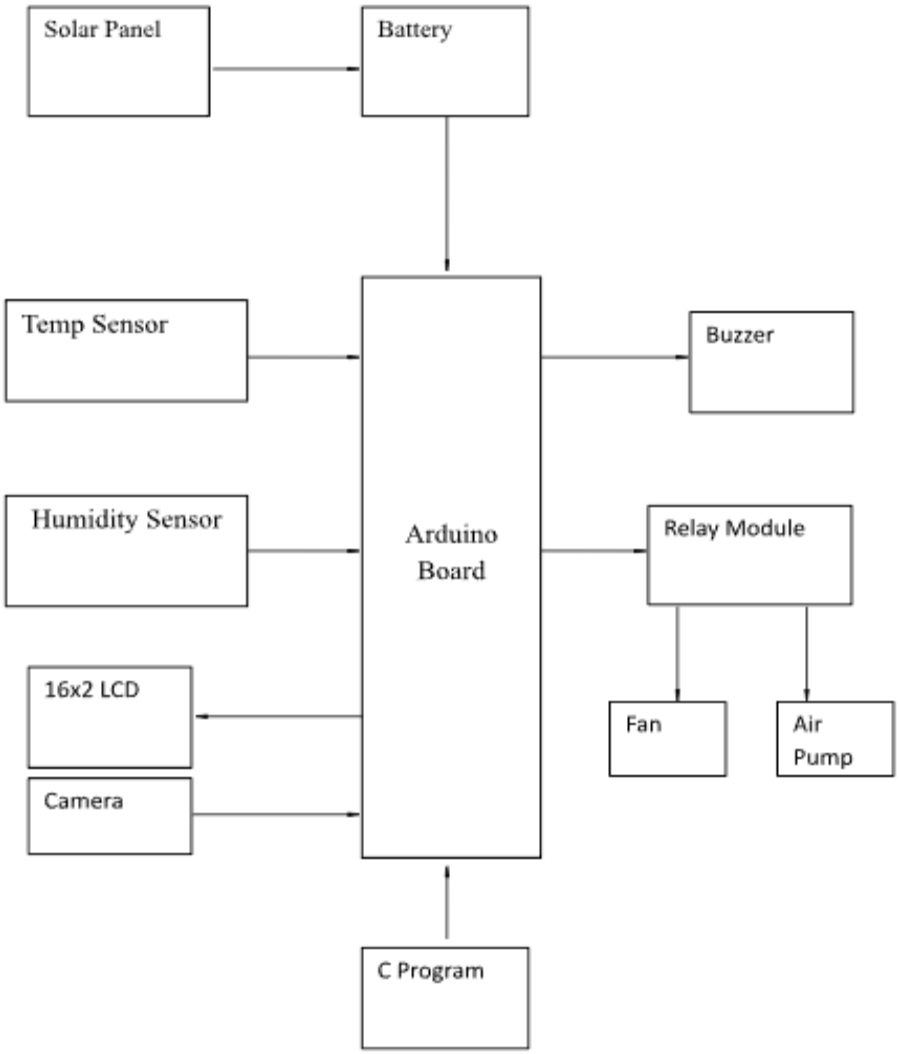
PROPOSED METHODOLOGY

5.1 Proposed Methodology

WORKING:

- Solar dryers are used to eliminate the moisture content from crops, vegetables, and fruits. The solar dryer consists of a box made up of easily available and cheap materials like cement, galvanized iron, brick, and plywood. The top surface of the dryer is covered by transparent single and double-layered sheets.
- The inside surface is colored black to absorb the incoming solar radiation. Since the box is insulated, the inside temperature of the box is raised. The air is ventilated through the small holes at the top of the box. As the inside air gets warm, it rises by the natural circulation process and removes the moisture from the fruits, vegetables, and crops placed in trays inside the box. To fill the vacuum, fresh air comes in through a forced draught process and the process continues.
- The Arduino controller automates the drying process based on preset values.
- The relay module controls the fan and LED lights for optimal drying conditions.
- Open the webcam application on your PC or smartphone to monitor the drying process through live video streaming.

5.2 Architecture/block diagram



5.3 Action Plan

Sr. no	Details Of activity	Planned Start date	Planned finish date	Name of Responsible Team Members
1	Problem Identification	15-12-23	21-12-23	Snehal Pawar Kajal Dhawade Sakshi Gawali Prashika Waghmare
2	Industrial Survey	22-12-23	25-12-23	Snehal Pawar Kajal Dhawade
3	Literature Review	25-12-23	29-12-23	Sakshi Gawali Prashika Waghmare
4	Selection of Capstone Project	29-12-23	5-01-24	Snehal Pawar Kajal Dhawade Sakshi Gawali Prashika Waghmare
5	Requirement Gathering	5-01-24	11-01-24	Snehal Pawar
6	Analyze Requirement	11-01-24	16-01-24	Snehal Pawar Sakshi Gawali
7	Define the scope Of project	16-01-24	20-01-24	Kajal Dhawade

8	Plan the Workflow of Project	20-01-24	11-01-24	Sakshi Gawali Prashika Waghmare
9	Completion of capstone project proposal	12-02-24	21-02-24	Snehal Pawar Kajal Dhawade Sakshi Gawali Prashika Waghmare
10	Data Gathering	22-02-24	25-02-24	Snehal Pawar Kajal Dhawade
11	Design Architecture	25-02-24	26-2-24	Sakshi Gawali Prashika Waghmare
12	Report Writing for work done in fifth semester and planning foe sixth semester work	26-02-24	27-02-24	Snehal Pawar Kajal Dhawade
13	Presentation and Defense for work Done in fifth Semester and planning for Sixth Semester Work	27-02-24	29-02-24	Snehal Pawar Kajal Dhawade Sakshi Gawali Prashika Waghmare

CHAPTER 6

PROGRAMMATIC CODE

```
#include <LiquidCrystal.h>

#include "DHT.h"

#define DHTPIN 8

#define DHTTYPE DHT11   // we are using the DHT11 sensor

LiquidCrystal lcd(12, 11, 5, 4, 3, 2);

DHT dht(DHTPIN, DHTTYPE);

const int TpotPin = A5;
const int HpotPin = A4;



void setup()
{


  pinMode(sol_heat, OUTPUT);
  pinMode(fan, OUTPUT);
  pinMode(TpotPin, INPUT);
  pinMode(HpotPin, INPUT);

  lcd.begin(16, 2); //16 by 2 character display

  dht.begin();
}



void loop()
{
  // Reading temperature or humidity
  int h = dht.readHumidity();

  int t = dht.readTemperature();  // Read temperature as Celsius (the default)
```

```
 int threshold_T = map(analogRead(TpotPin), 0, 1023, 0, 100); // Map potentiometer
value to temperature range

int threshold_H = map(analogRead(HpotPin), 0, 1023, 0, 100); // Map potentiometer
value to humidity range

  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("SV: ");
  lcd.print(threshold_T);  //printing temperarture to the LCD display
  lcd.print("'C");

  lcd.setCursor(10, 0);
  lcd.print(threshold_H);
  lcd.print("% H");


  lcd.setCursor(0, 1);
  lcd.print("PV: ");
  lcd.print(t);
  lcd.print("'C");

  lcd.setCursor(10, 1);
  lcd.print(h);
  lcd.print("% H");

delay(1000);

  if (t <= threshold_T)
  {
    digitalWrite(sol_heat, HIGH);

  }

  Else

  {

    digitalWrite(sol_heat, LOW);

  }
```

```
    if (h <= threshold_H)
     {
       digitalWrite(fan, LOW);
    else
     {
   digitalWrite(fan, HIGH);
  }
  }
```

CHAPTER 7

USER MANUAL

7.1 Installation and setup

- Plug the solar panel's output into the solar input jack on the main control board.
- Ensure correct polarity for a seamless operation.
- Place both the solar panel and the dryer in direct sunlight.
- Bright sunlight is essential for optimal performance.
- Connect the solar air heater pipe to the designated inlet point on the dryer box to facilitate efficient drying.
- Install a webcam application on your PC or smartphone.
- Connect the webcam to your device using the USB cable.
- Position the webcam for a clear view inside the dryer box.
- Enjoy live video streaming of the drying process.
- Turn on the main system by pressing the power button on the control board.
- Ensure the battery is fully charged for continuous operation.
- Utilize the two preset potentiometers on the right side of the LCD to set the desired temperature and humidity values for efficient drying.
- Contact us before operating the system for the first time.

7.2 Input to application

- In the input part of the application, we can operate all functions of

  Simulation with real-time drying in the field of solar dryer .

7.3 Output from application

- We get live streaming through camera.
- We get instant function response through solar dryer.

CHAPTER 8

ADVANTAGES & APPLICATION

Advantage:

- Energy efficiency
- Cost-effectiveness
- Low post-harvest
- Loss High product quality
- Environmental sustainability
- Small farmers
- Empowerment response
- Technology change and capacity building.

Application:

1. **Educational and Corporate Site Visits:** Gesture Sphere Explorer offers an immersive virtual experience replicating real-world sites, controllable via hand gestures. It provides a realistic alternative to physical travel, enabling efficient virtual site visits for institutes and companies. This enhances planning, collaboration, and resource savings.

2. **Realistic Structure Simulations:** Gesture Sphere Explorer allows realistic simulations of structures before physical construction, aiding in understanding functionality and feel. This streamlines the design process, identifies drawbacks and advantages, and leads to significant resource and cost savings by addressing structural attributes in virtual reality before real-world construction.

CHAPTER 9

CONCLUSION & FUTURE SCOPE

Conclusion:

Solar dryers provide safety, efficiency, and environmental protection for agricultural storage and drying applications, contributing to food security, poverty alleviation, and environmental management. Continued investment, research, and collaboration are essential to realize the full potential of solar drying technology and its positive impact on global food supply and development goals.

Future Scope:

The future scope for solar dryer projects is promising, with growing interest and investment in renewable energy technologies worldwide. As concerns about food security, climate change, and energy sustainability continue to rise, solar dryers offer an environmentally friendly and cost-effective solution for preserving agricultural produce and enhancing food security. Advancements in solar panel efficiency, battery storage technology, and control systems are expected to further improve the performance and reliability of solar dryers, making them more accessible and efficient for a wider range of applications. Additionally, integration with digital technologies such as IoT (Internet of Things) and remote monitoring systems holds the potential to enhance operational efficiency and enable data-driven decision-making. With increasing awareness of the benefits of renewable energy and sustainable agriculture, the future looks bright for the expansion and adoption of solar dryer projects globally.

CHAPTER 10

ACTUAL SCREEN LAYOUT

NEVA
Solar Air Heater
DHT11
Sensor
Exhaust Fan

CHAPTER 11

REFERENCES

11.1 Research papers:

We collected the data from internet Reference research papers:

- Tiwari, G.N. and Sodha, M.S. (2006). Before drying in the sun. Solar Energy, 80(10), 1043-1070.
- Kadam, D.M. and Goyal, R.K. (2012). Solar dryers: a review. Renewable and Sustainable Energy Reviews, 16(1), 2652-2670.
- Mujumdar, A.S. and Law, C.L. (2002). Dryers: trends and applications in post-harvest processing. Food and Bioprocess Technology, 84(2), 87-102.

ACHIEVEMENTS

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